

COM3502/4502/6502 SPEECH PROCESSING

Lecture 5

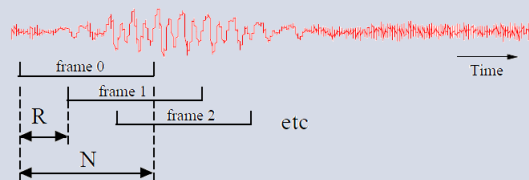
Waveform Processing

Block Processing

- After capture through a microphone and digitisation through the analogue-to-digital converter, an incoming speech signal becomes a sequence of quantised samples
- Digital signal processing is typically performed on a fixed-length sequence of samples called '**blocks**' or '**frames**'
 - e.g. in Pure Data the default '**block size**' is 64 samples (i.e. *1.45 msecs frame at the default 44.1 kHz sampling rate*)
- Because of the quasi-stationary nature of speech, the frame size is a compromise of ...
 - having *sufficient* data in a frame to make the required measurements
 - having *small enough* amount of data that the stationarity assumption is fulfilled
- It is also necessary to ensure that there are a sufficient number of frames to capture the *non-stationary* properties

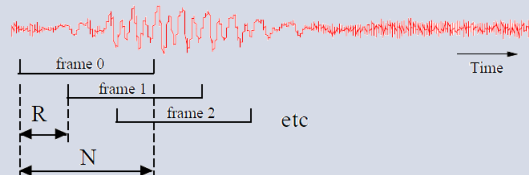
Block Processing

- To accommodate all these constraints, it is usual to use *overlapping* frames in speech processing
 - 'frame size' (N): number of samples per frame
 - 'frame shift' (R): number of samples between the start of successive frames
- Frame size is often expressed in time ...
 - NT seconds (where T is the sample period)
- Frame shift is often expressed as the 'frame rate' ...
 - $f_r = 1/RT$ frames per second (fps)



Block Processing

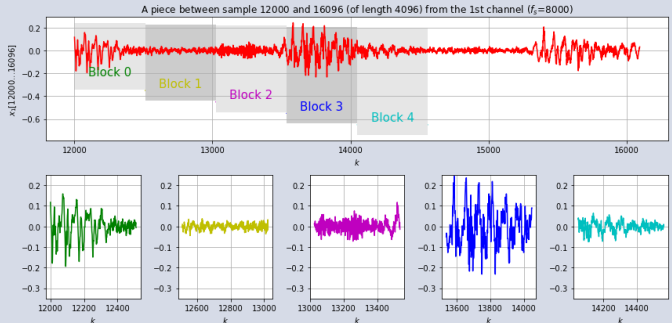
- In speech, it is usual to have ...
 - frame length (NT) ≈ 30 msecs
 - frame rate (f_r) ≈ 100 fps
- For example ...
 - sample rate (f_s) = 10 kHz (10,000 samples/sec)
 - sample period (T) = $1/f_s = 100$ μ secs/sample
 - frame size (N) = $NT/T = 0.03/0.0001 = 300$ samples
 - frame shift (R) = $1/f_r T = 1/(100 \times 0.0001) = 100$ samples
 - frame overlap ($N-R$) = $300-100 = 200$ samples (66%)



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Block Processing

- For example ...
 - sample rate (f_s) = 8 kHz (8,000 samples/sec)
 - sample period (T) = $1/f_s$ = 100 μsecs/sample
 - frame size (N) = 512 samples
 - No frame overlap



A piece between sample 12000 and 16096 (of length 4096) from the 1st channel ($f_s=8000$)

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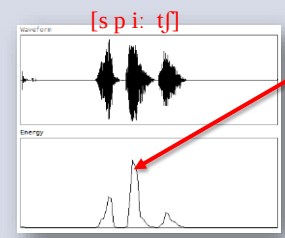
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Short-Time Energy

'Short-time energy'
= sum of the squares of the samples in one frame

$$E = \sum_{i=0}^{L_{B1}-1} s^2[iT_s]$$
$$= \sum_{i=0}^{L_{B1}-1} s^2[k]$$

Energy is large in voiced speech



E : short-time energy
 L_{B1} : block length
 $s[iT]$: sample no i of signal s

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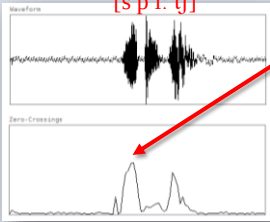
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Zero-Crossing Rate

'ZCR' = number of times the zero axis is crossed in one frame

ZCR is large in unvoiced speech

[s p i: t]



The figure shows two vertically stacked plots. The top plot, labeled 'Waveform', displays the amplitude of the word 'spit' over time. The bottom plot, labeled 'Zero-Crossings', shows the rate at which the waveform crosses the zero axis. A red arrow points from the 'ZCR is large in unvoiced speech' box to the high peaks in the Zero-Crossings plot that correspond to the unvoiced 's' and 't' sounds in the word 'spit'.



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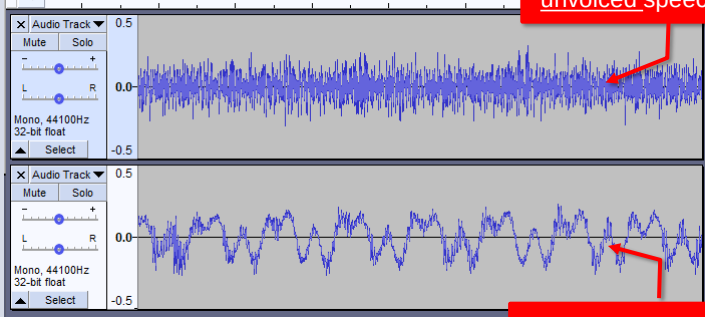
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Zero-Crossing Rate

'ZCR' = number of times the zero axis is crossed in one frame

ZCR is large in unvoiced speech



The figure displays two audio tracks side-by-side. The top track shows a noisy, high-frequency waveform characteristic of unvoiced speech, with a corresponding Zero-Crossing Rate (ZCR) plot showing many sharp, frequent peaks. The bottom track shows a smooth, periodic waveform characteristic of voiced speech, with a corresponding ZCR plot showing fewer, broader peaks. Red arrows point from the 'ZCR is large in unvoiced speech' box to the top ZCR plot, and from the 'ZCR is smaller in voiced speech' box to the bottom ZCR plot.

ZCR is smaller in voiced speech



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Speech/Non-Speech Detection

- In speech processing it is often useful to be able to detect when someone is speaking
- Accurate speech '**end-point detection**' is very difficult
- A simple '**speech/non-speech detector**' can be constructed using short-time energy *and* zero-crossing rate
 - energy is high in voiced speech
 - ZCR is high in unvoiced speech

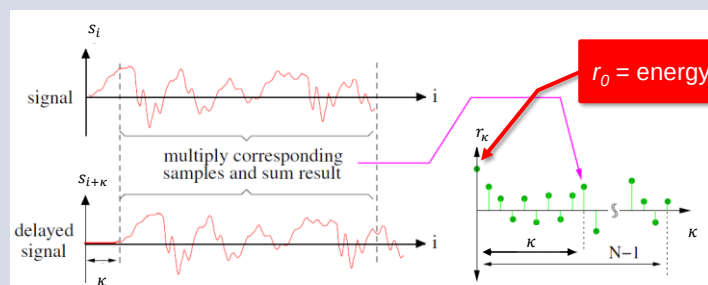
Autocorrelation Function

The '**autocorrelation function**' computes the correlation of a signal with itself (as a function of time)

2 different ways of writing the same equation

$$r_{\kappa} = \sum_{i=0}^{N-\kappa-1} s_i s_{i+\kappa}$$

$$r[\kappa] = \sum_{i=0}^L s[i] s[i + \kappa]$$



Autocorrelation Function

- The autocorrelation function (**ACF**) emphasises periodicity
- ACF is the basis for many spectrum analysis methods
- Short-time ACF (**STACF**) is the basis for many '**pitch detectors**' (*fundamental frequency estimators*)
- ACF is fairly expensive to compute (*because there is an inner loop running for every data sample*)
- STACF is often combined with ZCR to construct a '**voiced/unvoiced detector**'

Covariance & Correlation

The '**covariance**' between two discrete-time signals $x[k]$ and $y[k]$ is ...

$$c_{xy}[k_1, k_2] = E\{(x[k_1] - \mu_x) \cdot (y[k_2] - \mu_y)\}$$

The '**auto-covariance**' between a discrete-time signals $x[k]$ the same signal $x[k]$ is ...

$$c_{xx}[k_1, k_2] = E\{(x[k_1] - \mu) \cdot (x[k_2] - \mu)\}$$

Assuming '**wide-sense stationarity**' (WSS) the covariance only depends on the difference $\kappa = k_2 - k_1$ between the considered sample indices, i.e. $k_2 = k_1 + \kappa$

$$c_{xx}[\kappa] = E\{(x[k] - \mu) \cdot (x[k + \kappa] - \mu)\}$$

Covariance & Correlation

The '**(auto)-correlation**' between two discrete-time signals $x[k]$ and $y[k]$ is ...

$$r_{xx}[\kappa] = \frac{c_{xx}[\kappa]}{\sigma_x \sigma_x} \quad r_{xy}[\kappa] = \frac{c_{xy}[\kappa]}{\sigma_x \sigma_y}$$

A simple '**covariance estimator**' for two discrete-time signals $x[k]$ and $y[k]$ replaces the expectation $E\{..\}$ by a sum over the samples...

$$\hat{c}_{xx}[\kappa] = \frac{1}{L} \sum_{k=1}^L (x[k] - \mu)(x[k + \kappa] - \mu)$$

With '**sample mean**' and '**variance**' estimate

$$\mu_x = \frac{1}{L} \sum_{i=1}^L x[i] \quad \hat{\sigma}_x^2 = \frac{1}{L} \sum_{i=1}^L (x[i] - \mu_x)$$

Correlation

A simple '**correlation estimator**' for two discrete-time signals $x[k]$ and $y[k]$ can thus be defined as...

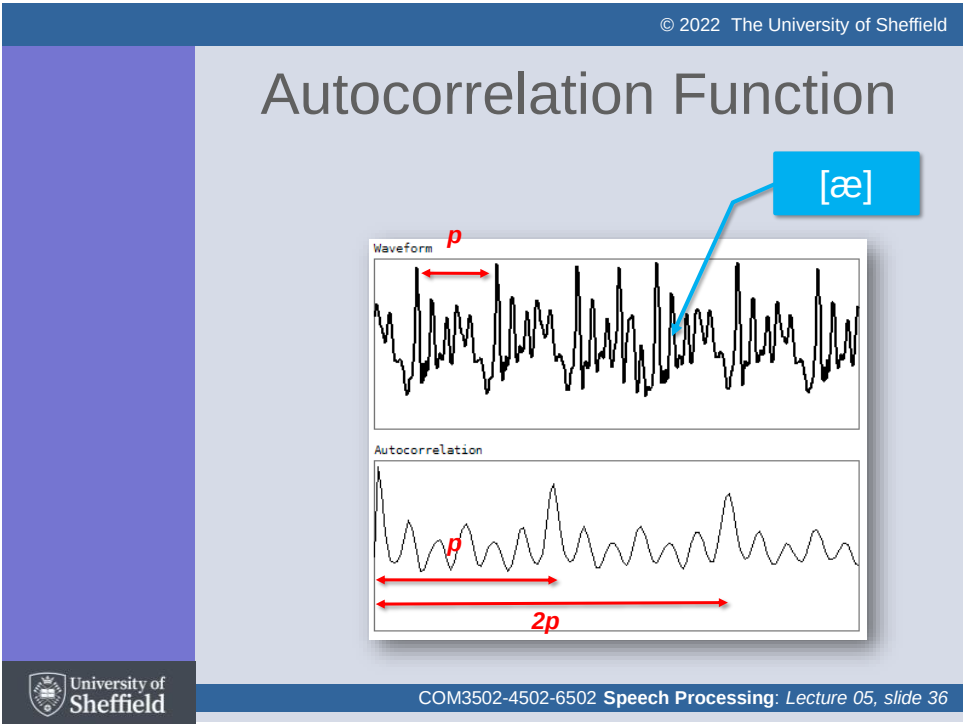
$$\hat{r}_{xy} = \frac{\sum_{k=1}^L (x[k] - \hat{\mu}_x)(y[k] - \hat{\mu}_y)}{L \cdot \hat{\sigma}_x \hat{\sigma}_y}$$

With '**zero-mean**' $\hat{\mu}_x = 0, \hat{\mu}_y = 0$ and standard deviation of 1, i.e. $\hat{\sigma}_x = 1, \hat{\sigma}_y = 1$ this simplifies to...

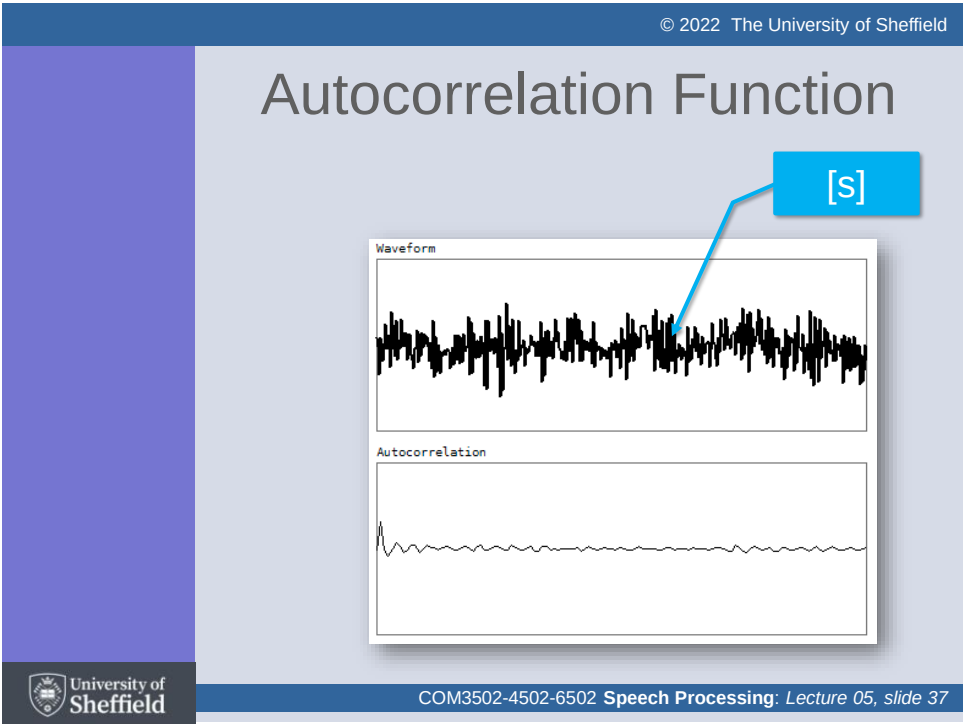
$$\hat{r}_{xy} = \frac{1}{L} \sum_{k=1}^L x[k]y[k]$$

Compare to
'simple' definition
from lecture
'Introduction'

$$r_{xy}[\kappa] = E\{x[k]y[k + \kappa]\}$$



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This slide set has covered ...

- Processing blocks/frames
- Processing samples
- Short-time energy
- Zero crossing rate
- Speech/non-speech detection
- (Auto)correlation and covariance

Any Questions ?

*Raise during lecture or post in the Blackboard
Discussion group*



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Next slide set ...

Frequency Analysis

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